

PROJECT
BACKBENCHERS

GAME DESIGN DOCUMENT

Prototype / Vertical Slice Specification

Project Title	Backbenchers
Genre	Visual Narrative RPG
Setting	Malaysian secondary school
Platforms	Mobile + PC
Orientation	Horizontal
Prototype Duration	3 Chapters
Prototype Goal	Adapt to a new school, form relationships, and pass the first major exam
Long-Term Character Goal	Eventually pass SPM
Tone	Comedy + coming-of-age; lighthearted and socially open-minded
Document Status	Working GDD — systems-first prototype
Version / Date	v0.2.2 — Updated until 7.Minigames, Action Slots and Activity Phase (Minigames remained unchanged, need more time to look over) 24/8/26

DOCUMENT RULE

This GDD separates confirmed design decisions from working assumptions and unresolved areas. Anything marked TBD is intentionally left open for later design/testing rather than being treated as final.

1. Executive Summary

Backbenchers is a strictly vertical mobile and PC visual novel / social RPG set in a Malaysian secondary school. The player takes the role of a Form 3 transfer student entering a new college/boarding school. The long-term character goal is to eventually pass SPM; the current prototype focuses on only the first 3 Chapters, ending with the protagonist's first major exam.

The core experience is now structured around an episodic, narrative-first format. Across three chapters, the player experiences authored school-life situations that can happen anywhere within the boarding school rather than being confined to a fixed timetable or daily routine. Story beats lead naturally into dialogue choices, D20 checks, character attributes, Best Friend interactions, and a variety of short subject-based minigames. Between major narrative segments, players receive a limited number of Action Slots that can be spent hanging out with friends, doing chores for money, working out to improve Dexterity, or studying to improve Intelligence and practice subject minigames. Money can also be spent on food and other preparation items that provide temporary gameplay buffs.

The prototype is intended to prove that a dialogue-first school RPG can feel mechanically active and varied without relying on combat, a large explorable hub, or a rigid day-to-day simulation. Its focus is on giving the narrative freedom to drive progression while RPG systems allow the player to shape how their character approaches problems. Academic minigame performance builds toward a final exam gauntlet in Chapter 3, while Charisma, Intelligence, Dexterity, Best Friend approval, money, and previous choices influence how the story unfolds. The horizontal presentation also provides more room for character interactions, stylized UI, minigames, and environmental composition.

Project Vision

[LOCKED] Create a light hearted replayable coming-of-age RPG where ordinary Malaysian boarding-school life is treated with the reactivity, choice, and mechanical consequence of a tabletop role-playing game.

[LOCKED] Across three episodic chapters, the player should feel that their character build, Best Friend, Action Slot choices, spending, preparation, minigame performance, and moment-to-moment reactions create a personal version of the same core story, leading toward the Chapter 3 final exam gauntlet.

Prototype Success Condition

The prototype succeeds if its three-chapter run demonstrates that the player's build and choices meaningfully alter school situations, relationships, available dialogue responses, preparation activities, academic performance, and the final exam outcome without requiring a large open world, rigid day-to-day simulation, or conventional quest structure.

2. Game Overview

Category	Current Design
Player Character	Form 3 transfer student from another school
Primary Goal	Adapt to the school and pass the first major exam within the 3 Chapter
Long-Term Motivation	Eventually pass SPM
Core Genre	Visual Narrative RPG
Tone	Comedy and coming-of-age; lighthearted, socially open-minded
Perspective / Screen	Horizontal
Core Input	Dialogue selection, tapping/clicking, occasional gyro/camera movement, minigames
Primary Resource	Action Slots
Secondary Resource	Money (RM)
Companions	3 total Best Friends; 1 active at a time
Failure Model	Fail-forward; failed rolls create different outcomes rather than routine Game Overs
Save Model	Manual saving/reloading allowed; save-scumming is acceptable

Primary Audience

[WORKING] Players who enjoy character-driven visual novels, social RPGs, choice-and-consequence games, and lighter tabletop-inspired systems. Exact age rating and market positioning are TBD.

Non-Goals for the Prototype

- No traditional combat system is required.
- No large walkable school hub is required.
- No conventional side-quest checklist is required.
- No large survival/resource simulation (energy, hunger, stress, etc.).
- No requirement that every small choice permanently alter the ending.

3. Design Pillars

3.1 Narrative Comes First

Backbenchers is an episodic, narrative-driven RPG. Each chapter follows meaningful school-life situations rather than simulating every day or forcing the story into a fixed timetable. Gameplay systems should support the narrative, not restrict where or when story events are allowed to happen.

3.2 Build-Driven Role-Playing

The player's Class, Charisma, Intelligence, and Dexterity should create different ways of approaching the same situation. A high-Charisma student should experience situations differently from a high-Intelligence or high-Dexterity student.

3.3 Reactive Choices, Personal Outcomes

The player is frequently placed into situations and asked to react rather than search for quests or content. Dialogue choices, D20 checks, Best Friend interactions, previous decisions, and character attributes should make the player's version of the story feel personal even when the main narrative remains shared. Not every decision needs a permanent consequence, but every meaningful decision should produce an immediate response from the game.

3.4 Failure Is Content

Failure should create new situations rather than simply stop progression.

3.5 3D With Purpose, Not Obligation

Gyro/parallax movement is allowed to be presentation-only most of the time. Selected Investigation Scenes turn the environment into gameplay. This keeps the feature distinctive without forcing every background to become a complex interactive level.

3.6 Limited Freedom Creates Character

Between major narrative segments, Action Slots allow the player to decide what kind of student they want to become. Money can then be spent on food, items, or temporary advantages. The player cannot improve everything at once, so their priorities gradually define their build and relationships.

4. Player Experience and Gameplay Loops

4.1 Primary Gameplay Loop — Narrative Role-Playing

The primary gameplay loop is the player's moment-to-moment experience throughout each chapter.

Experience Situation → Observe → Respond → Resolve → Consequence → Continue Story

→ **Experience Situation**

The narrative places the player into a school-life situation, conversation, conflict, or event.

→ **Observe**

The player reads the dialogue, character behaviour, available information, and relevant context.

→ **Respond**

The player reacts through:

- ◆ Dialogue choices
- ◆ Attribute-specific options
- ◆ Best Friend options
- ◆ Contextual actions
- ◆ Items
- ◆ Occasional minigames

→ **Resolve**

The selected action either resolves automatically or triggers a D20 check against a contextual Difficulty Class (DC).

→ **Consequence**

The result may affect:

- ◆ Dialogue
- ◆ Best Friend approval
- ◆ Money
- ◆ Items
- ◆ Character stats
- ◆ Story flags
- ◆ Immediate circumstances
- ◆ Later scenes

→ **Continue Story**

The chapter progresses into the next narrative situation.

4.2 Secondary Gameplay Loop — Free-Time Preparation

At selected points throughout a chapter, the player receives a limited number of **Action Slots**.

These allow the player to deliberately shape their character between mandatory narrative events.

Receive Action Slots → Choose Activity → Gain Benefit → Spend Resources → Return to Story

Activity	Primary Reward	Additional Benefit
Hang Out	Increase Charisma +2	Increase Best Friend approval and experience character scenes
Chores / Favours	Gain Money	Provides spending power for food, items, and preparation
Work Out	Increase Dexterity+2	Improves physical and Acrobatics-related options
Study	Increase Intelligence+2	Allows practice of a selected academic minigame

The player cannot complete every available activity.

Action Slots therefore create build differentiation and encourage the player to decide what matters most to their version of the protagonist.

Money earned through chores can be spent on food, items, and temporary buffs without necessarily consuming an Action Slot.

After preparation is complete, the player returns to the chapter's main narrative.

4.3 Tertiary Gameplay Loop — Academic Challenges

Academic minigames provide short mechanical challenges throughout the narrative and build toward the Chapter 3 Final Exam Gauntlet.

Encounter Academic Challenge → Play Minigame → Receive Score → Build Subject Readiness → Prepare for Final Exam

Current prototype minigames:

Subject	Minigame	Core Skill
Math	Calc Combo	Mental calculation and equation construction
Science	Reaction Chain	Scientific recognition and association
English	Error Hunt	Grammar recognition, reading speed, and accuracy

Academic challenges may appear wherever appropriate within the narrative and are **not tied to a fixed daily timetable**.

Performance contributes to separate Subject Scores for:

- Math
- English
- Science

These scores represent the player's academic preparation and contribute to the Chapter 3 Final Exam result.

Final Exam

The Final Exam Gauntlet uses the hardest versions of all three minigames.

Final Subject Result = 70% Final Exam Performance + 30% Subject Score

The player must also achieve at least **50% on the actual exam portion**, ensuring that preparation helps without removing the challenge of the final exam.

Loop Hierarchy

Primary Loop — Narrative Role-Playing

The main experience. Story situations, dialogue, decisions, rolls, and consequences.

↓ supports / is influenced by

Secondary Loop — Free-Time Preparation

The player develops stats, relationships, money, and preparation between story beats.

↓ supports

Tertiary Loop — Academic Challenges

Short skill-based minigames build academic performance toward the Final Exam Gauntlet.

All supporting systems should ultimately feed back into the **Primary Narrative Loop**.

4.5 Flowchart

<https://canva.link/nraawdr2mqy67In>

5. Character Creation, Classes and Attributes

5.1 Character Creation

The protagonist is customizable enough to support player ownership without requiring an expensive full avatar creator.

- Name
- Gender
- A small set of appearance options/presets
- Starting Class
- Attribute allocation

5.2 Base Attributes

The prototype uses only three core attributes. Each begins at 5. The player receives 12 additional attribute points to allocate during character creation, after which Class modifiers are applied.

Attribute	Checks / Uses
Charisma (CHA)	Performance, Persuasion
Intelligence (INT)	Awareness, Deception, Investigation
Dexterity (DEX)	Strength, Acrobatics

[LOCKED] Do not add additional core attributes during the prototype unless testing shows a clear need.

[TBD] Maximum attribute cap during character creation. A pre-Class cap of 10 is a current balancing candidate, not yet locked.

5.3 Starting Classes

Class	Starter	Playstyle / Trade-off
Academician	CHA(6), INT(12), DEX(8)	Strong analytical/investigation build with weaker social presence
Athlete	CHA(8), INT(6), DEX(12)	Strong physical/reactive build with weaker analytical checks
Class Clown	CHA(12), INT(8), DEX(6)	Extreme social specialist with major academic and physical weaknesses

//update balik

Classes are starting identities rather than rigid professions. The player can use the 4 allocation points to compensate for weaknesses or specialize further.

6. Dialogue, Checks and D20 Resolution

6.1 Dialogue Philosophy

Dialogue is the primary gameplay system. Choices may be freely available, attribute-gated, context-gated, Best-Friend-gated, or tied to information found in an Investigation Scene.

Not every dialogue choice must permanently branch the whole narrative. Small reactive school incidents are allowed to resolve locally and reward items, money, approval, buffs, or character flavour.

6.2 D20 Check Rule

[LOCKED] All random checks use a D20 rolled against a preset contextual DC.

Recommended prototype formula: D20 + Attribute Modifier versus DC.

Raw Attribute	Modifier
2(lowest cap)	-1
4	-1
6	+0
8	+0
10	+1
12	+1
14	+1
16	+1
18	+1
20	+1

The DC is authored per context rather than generated from a universal difficulty formula. This lets writers/designers control how risky each school situation should feel.

[WORKING] Natural 1 = Critical Failure and natural 20 = Critical Success is the preferred prototype rule. Ordinary rolls below the DC are Failure; rolls meeting/exceeding the DC are Success. Confirm in balance testing.

6.3 Four Outcome States

State	Intent
Critical Failure	Worst or most chaotic authored outcome; still usually progresses the scene
Failure	The chosen approach does not work; creates a different consequence
Success	The chosen approach works as intended
Critical Success	Works exceptionally well and may grant bonus information/reward/dialogue

6.4 Save/Reload Policy

[LOCKED] Save-scumming is allowed. The prototype will not spend production effort trying to prevent players from reloading rolls.

7.0 Minigames, Action Slots and Activity Phase

This section defines the supporting gameplay systems used throughout the three-chapter structure of **Backbenchers**.

The narrative remains the primary experience. Minigames and Activity Phases appear at authored points throughout each chapter to provide mechanical variety, character progression, preparation, and academic progression.

7.1 Academic Minigame System

The prototype contains three short academic minigames. They appear at authored points within the narrative and are designed as fast gameplay breaks rather than a persistent academic simulation.

Subject	Minigame	Core Skill	Main Interaction
Mathematics	Calc Combo	Mental calculation, speed, problem solving	Build equations that reach target numbers
Science	Lab Link	Visual logic, pattern matching, spatial reasoning	Arrange plant-themed hex tiles by matching connectors and required quantities
English	Error Hunt	Grammar recognition, reading speed, accuracy	Identify the incorrect word in short sentences

Shared Prototype Format

Setting	Prototype Rule
Round Duration	60 seconds
Base Reward	+10 points per completed challenge
Score Limit	None - score is uncapped
Primary Goal	Exceed 100 points before time expires
Difficulty	One standardized prototype difficulty
Progression	No persistent Subject Score, readiness, grade, or Final Exam contribution
Replay Goal	Improve personal score through faster and more accurate play

All three minigames follow the same high-level rhythm: **Challenge -> Solve -> +10 -> Next Challenge -> Repeat Until 60 Seconds End**

7.2 Mathematics - Calc Combo

Calc Combo is a rapid mental-arithmetic game. The player creates a valid equation for the displayed target, earns 10 points, and immediately receives a new target. The challenge is to solve as many equations as possible within 60 seconds.

Category	Prototype Design
Goal	Create valid equations that equal the displayed target number
Example	TARGET: 24 -> $6 \times 4 = 24$
Main Input	Calculator-style number and operator buttons
Core Skill	Mental calculation + speed
Round Length	60 seconds
Scoring	+10 points per correct equation; uncapped total
Target	Exceed 100 points
Mistake	Incorrect equation earns no points; player immediately continues
Replay Variety	Random target values, number combinations, and valid equation possibilities
Important Rule	Every generated target must have at least one valid solution
Difficulty	One standardized prototype difficulty

7.2.1 Prototype Settings

Setting	Working Value
Target Number Range	Approximately 5-50
Number Pool	1-9
Operators	+, -, x, /
Equation Length	Short equations; approximately 3-5 inputs
Timer	60 seconds
Minimum Valid Solutions	At least 1
Decimals	Disabled
Fractions	Disabled
Brackets	Disabled

7.2.2 Round Flow

Generate Target -> Build Equation -> Submit -> +10 / No Score -> New Target -> Repeat -> Final Score

7.2.3 Future Difficulty Scaling

[FUTURE DEVELOPMENT] The prototype does not implement multiple difficulty levels. The following variables may be used if the minigame is expanded later.

Variable	Possible Future Scaling
Target Range	Increase maximum target number
Operators	Restrict or expand operator combinations
Equation Length	Require longer equations
Number Pool	Increase available values
Timer	Reduce available time
Decimals / Fractions	Introduce more complex number types
Brackets	Introduce order-of-operations puzzles

7.3 Science - Lab Link

Lab Link is a fast hexagonal matching game themed strictly around plants, water, and sunlight. The player completes a series of small logic puzzles by using the required number of each tile type while matching symbols on touching hexagon edges.

The science content is intentionally simple. The challenge comes from visual logic and speed rather than detailed scientific knowledge.

Category	Prototype Design
Theme	Plants and the basic requirements for plant growth
Tile Types	Plant / Water / Sunlight
Goal	Complete a small hex board using the required tile quantities while matching adjacent connector symbols
Main Input	Select or drag hexagonal tiles into empty spaces
Core Skill	Visual logic + spatial reasoning + pattern matching
Round Length	60 seconds
Scoring	+10 points per completed board; uncapped total
Target	Exceed 100 points
Mistake	Invalid placement is rejected; no score is awarded until the board is solved
Rotation	Disabled
Replay Variety	Tile requirements, tile pool, connector arrangement, and small board layout
Difficulty	One standardized prototype difficulty

7.3.1 Core Puzzle Rules

Each micro-puzzle provides a required amount of Plant, Water, and Sunlight tiles, along with a small fixed pool of available pieces. The player must satisfy both the quantity requirement and the connector rules.

Example Requirement	Example Available Tiles
Plant x1 Water x1 Sunlight x1	Plant x2 Water x1 Sunlight x1

If the board contains three spaces, the player chooses the correct three tiles from the available pool and places them so that all touching connector symbols match.

7.3.2 Connector System

Connector symbols are purely puzzle mechanics. They do not represent additional scientific concepts. Touching edges are valid only when their symbols match.

Connector Example	Result
Circle <-> Circle	Valid
Triangle <-> Triangle	Valid
Circle <-> Square	Invalid

The prototype should use only three visually distinct connector types, such as Circle, Triangle, and Square, so players can read the board instantly.

7.3.3 Prototype Settings

Setting	Working Value
Board Spaces	Approximately 3-4 per micro-puzzle
Available Tiles	Approximately 4-5
Tile Categories	3: Plant / Water / Sunlight
Extra / Unused Tiles	Approximately 1
Connector Types	3
Timer	60 seconds for the full minigame session
Tile Rotation	Disabled
Valid Solutions	At least 1 per generated board
Pacing Target	Each micro-puzzle should be simple enough that a skilled player can regularly exceed 100 total points

7.3.4 Puzzle Generation

Generate Valid Board -> Set Required Tile Counts -> Add Extra Tile -> Shuffle Pool -> Solve -> +10 -> Generate Next Board

The system must generate a valid solution before shuffling the tile pool. This prevents procedurally generated puzzles from becoming impossible.

7.3.5 Future Difficulty Scaling

[FUTURE DEVELOPMENT] Difficulty should increase through puzzle complexity rather than more obscure science knowledge.

Variable	Possible Future Scaling
Board Size	Increase the number of hex spaces
Tile Pool	Add more unused or decoy tiles
Connector Types	Increase beyond three symbols
Required Quantities	Use more specific Plant / Water / Sunlight combinations
Timer	Reduce available time
Pre-Placed Tiles	Add fixed tiles that constrain solutions
Tile Rotation	Allow or require rotation
Board Shape	Introduce more complex layouts

7.4 English - Error Hunt

Error Hunt is a rapid grammar-recognition game. One short sentence appears at a time with a single controlled language error. The player taps the incorrect word, earns 10 points, and immediately receives another sentence.

Category	Prototype Design
Goal	Identify the incorrect word as quickly and accurately as possible
Example	She go to school yesterday. -> tap "go"
Main Input	Tap the incorrect word
Core Skill	Reading speed + grammar recognition + accuracy
Round Length	60 seconds
Scoring	+10 points per correctly identified error; uncapped total
Target	Exceed 100 points
Mistake	Incorrect tap earns no points; sentence remains active or refreshes according to final implementation
Replay Variety	Random validated sentences and controlled error types
Important Rule	Sentences must be authored and validated; arbitrary sentence generation is not used
Correction Step	Not required in the initial prototype; correction may be shown automatically as feedback
Difficulty	One standardized prototype difficulty

7.4.1 Error Content Pool

Error Type	Incorrect Example	Correct Form
Verb Tense	She go yesterday.	She went yesterday.
Subject-Verb Agreement	He play football.	He plays football.
Singular / Plural	There are three book.	There are three books.
Article	She bought a umbrella.	She bought an umbrella.
Pronoun	Sarah forgot his bag.	Sarah forgot her bag.
Preposition	He arrived in school at 7 AM.	He arrived at school at 7 AM.
Spelling	The view was beautifull.	The view was beautiful.
Capitalization	we went home.	We went home.
Punctuation	Where are you going.	Where are you going?
Word Choice	I borrowed him my pencil.	I lent him my pencil.

7.4.2 Prototype Sentence Rules

Setting	Working Value
Errors per Sentence	1
Sentence Length	Short to medium
Sentence Selection	Random from validated pool
Error Selection	Only controlled errors valid for the selected sentence
Timer	60 seconds for the full minigame session
Correction Selection	Not required initially
False Alarm / No-Error Sentences	Disabled
Incorrect Tap Penalty	TBD; should remain small enough to preserve fast pacing

Example Procedural Variant

Validated: Aina finished her homework yesterday.

Presented: Aina finish her homework yesterday.

Player action: Tap "finish" -> +10 points -> next sentence.

7.4.3 Round Flow

**Display Sentence -> Find Error -> Tap Word -> +10 / No Score -> Immediate Feedback
-> Next Sentence -> Repeat -> Final Score**

7.4.4 Future Difficulty Scaling

[FUTURE DEVELOPMENT] The following options may be explored after the core prototype is validated.

Variable	Possible Future Scaling
Errors per Sentence	Increase from one to multiple errors
Sentence Length	Introduce longer sentences
Error Similarity	Use subtler grammar errors
False Alarms	Add correct sentences requiring a NO ERROR response
Correction Selection	Require the player to choose the correct replacement
Timer	Reduce available time
Incorrect Tap Penalty	Increase the cost of inaccurate tapping
Error Types	Add more advanced grammar categories

7.5 Shared Minigame Scoring

All three minigames use the same simple arcade-style scoring philosophy. Scores belong only to the current minigame session and are not converted into grades or persistent academic progression.

Setting	Rule
Correct Completion	+10 points
Incorrect Attempt	0 points; exact time/feedback penalty depends on minigame
Score Cap	None
Round Duration	60 seconds
Target	Exceed 100 points
Meaning of 100	10 successfully completed challenges
Meaning of >100	Player completed at least 11 challenges in the round
Persistent Academic Score	None
Final Exam Contribution	None

Design Target Each minigame must be tuned so that 100 points is achievable, while exceeding 100 requires competent, fast play. The scoring system should encourage immediate replay through personal score improvement rather than progression rewards.

7.5.1 Completion Definition

Minigame	A Completed Challenge Worth +10
Calc Combo	Submit one valid equation that reaches the target number
Lab Link	Complete one valid micro-board satisfying both tile quantities and connector rules
Error Hunt	Correctly identify the single error in one sentence

7.6 Prototype Scope and Future Expansion

The academic minigames are designed as short gameplay bursts inside a Visual Narrative RPG. Their purpose is to vary interaction and give the player a repeatable skill challenge without creating a large academic simulation system.

Current Prototype Priorities

- Instantly understandable rules
- Approximately 60-second sessions
- Fast reset between individual challenges
- Uncapped arcade-style scores
- Mechanically distinct subjects
- Repeatability through procedural or controlled content variation
- Simple implementation suitable for the prototype schedule

Not Required for the Current Prototype

- Multiple difficulty modes
- Persistent Subject Scores or Subject Readiness
- Final Exam systems
- Academic grades or pass/fail progression
- Curriculum-heavy science questions
- Complex correction stages in Error Hunt
- Complex tile rotation or large Lab Link boards

Future development may expand difficulty, content pools, modifiers, rankings, or progression only after the three core minigames are proven fun and readable at their simplest form.

7.7 Activity Phase

At selected points throughout each chapter, the narrative temporarily gives the player control over how to spend their free time.

These sections are called **Activity Phases**.

[LOCKED] Each chapter provides one Activity Phase with 3 Action Slots.

Activity Phases provide a controlled opportunity for character progression and preparation without forcing the narrative into a daily or weekly schedule.

Activity Phase Loop

Receive 3 Action Slots → Choose Activities → Gain Benefits → Optional Purchases → Starts Chapter Mandatory Minigame

The exact placement of the Activity Phase within each chapter is determined by the narrative.

7.7.1 Action Slots

Action Slots represent the player's limited free time during a chapter.

The player cannot perform every available activity, creating an opportunity cost between:

- Improving Charisma
- Improving Intelligence
- Improving Dexterity
- Earning Money
- Preparing academically

7.7.2 Activity Types

Activity	Action Slot Cost	Cost	Primary Reward / Function
Hang Out	1	—	Increase Charisma and experience optional character interactions
Chores / Favours	1	—	Earn Money
Work Out	1	—	Increase Dexterity

Study	1	—	Increase Intelligence and replay a selected academic minigame for practice
Purchase Food / Item	0	Money	Gain optional temporary buffs or preparation items

Activities may contain short authored dialogue or presentation where appropriate, but should remain simple enough to maintain prototype scope.

Study Rule

Studying allows the player to practise a selected academic minigame while increasing Intelligence.

[WORKING] Study-mode minigames should provide reduced or no direct Subject Score compared with mandatory academic challenges to prevent Study from becoming the dominant progression strategy.

Purchase Rule

Purchasing food or items requires **Money but does not consume an Action Slot**.

The player may therefore spend Money during an Activity Phase without sacrificing one of their three character-progression activities.

7.8 Money, Food and Preparation

[LOW PRIORITY]

Money, food, and preparation are supporting systems rather than core requirements for the prototype.

They should only receive significant development time after the narrative, dialogue, D20, Action Slot, attribute, and minigame systems are functioning.

7.8.1 Money

Money is primarily earned through **Chores / Favours**.

Completing Chores / Favours:

- Costs **1 Action Slot**
- Rewards Money

Money may then be spent during the Activity Phase without consuming additional Action Slots.

[TBD] Exact Money rewards and economy values.

7.8.2 Food and Items

Food and preparation items provide optional temporary advantages.

Possible effects include:

- Additional minigame time
- One forgiven mistake
- Reduced penalty for an incorrect input
- Small score bonus
- Context-specific temporary advantage

Food should support preparation without becoming a mandatory maintenance system.

[LOW PRIORITY / TBD]

- Food catalogue
- Prices
- Buff values
- Buff duration
- Stacking rules
- Storage / inventory behaviour

If development time becomes limited, the food and item system may be simplified or removed without affecting the primary prototype experience.

7.9 Companion Story Choice

The companion system has been significantly reduced to maintain development scope.

[LOCKED] The prototype contains three main companion characters.

The player chooses one companion whose perspective and personal storyline they experience more closely.

The companion choice primarily determines:

- Character-focused scenes
- Additional dialogue
- Narrative perspective
- Companion-specific story content

The system exists primarily to support **narrative variation and replayability**.

The prototype does **not** require:

- Companion stat buffs
- Passive abilities
- Approval meters
- Companion skill trees
- Complex switching systems
- Companion-specific mechanical progression

Companions should therefore be treated primarily as **narrative routes**, not gameplay equipment.

[TBD] Exact point at which the companion route is selected and whether the choice becomes permanently locked for the remainder of the prototype.

7.10 Balance Notes / Adjustable Values

System	Values Most Likely to Change During Testing
Calc Combo	Target range, operators, equation length, timer, input pool size
Reaction Chain	Answer count, timer, distractor similarity, category mixing, Chain reward
Error Hunt	Error count, sentence length, timer, false-alarm chance, incorrect-tap penalty
Shared Scoring	Accuracy / speed / combo weights and score thresholds
Subject Progression	Average Subject Score reward, Final Exam weighting, pass threshold
Action Slots	Locked at 3 per chapter for the prototype; activity rewards may still change
Hang Out	Charisma reward and amount of optional narrative content
Work Out	Dexterity reward
Study	Intelligence reward and practice-mode Subject Score rules
Chores / Favours	Money reward
Food / Items	Low priority — prices, buffs, duration and availability
Companion Routes	Amount of route-specific narrative content and selection point

All unmarked numeric values should remain adjustable until tested in the playable prototype.

8. World and Narrative Direction

8.1 Premise

The protagonist is a Form 3 transfer student arriving from another school. They must learn the culture of their new school, make friends, adapt to expectations, and prepare for an important exam in 30 days. SPM is mentioned as the larger long-term goal beyond the prototype.

8.2 Tone

The story should primarily be a comedy and coming-of-age experience. It can address relationships, school culture, identity, mistakes, pressure, friendship, and social differences, but the prototype should remain lighthearted overall rather than becoming a heavy drama.

[LOCKED] The tone should be socially open-minded/liberal in its character treatment and themes without turning the prototype into a political lecture.

8.3 Story Structure

[TBD] Main plot, major characters, Best Friend identities, school name/location, day-by-day story beats, exam narrative payoff, and endings will be discussed separately.

8.4 MoodBoard

<https://canva.link/q605lmqlq1qmj5h> (Canva Link)

9. Presentation and Technical Direction

9.1 Visual Direction

The UI direction takes inspiration from Persona 5: bold, graphic, energetic, and expressive. The goal is not to copy Persona 5, but to achieve similarly strong feedback for dialogue, rolls, choices, stats, and character reactions within Backbenchers' own Malaysian school identity.

Character art, UI/UX, and 3D/background work may be handled by different artists to allow specialization.

9.2 Platform / Orientation

[LOCKED] Mobile and PC, strictly vertical.

PC controls are considered straightforward: mouse/touch-equivalent input for dialogue and interaction, with mouse movement or click-drag replacing device tilt where required.

9.3 Minigames

[TBD] Math, English, Science, and any reaction minigames will be designed later. The fallback for scope pressure is to use simple reusable minigame templates rather than custom mechanics for every event.

9.4 UI Screens

[TBD] Detailed screen list, HUD layout, character sheet, dice UI, Best Friend UI, recess shop, Action Slot UI, save/load flow, and settings will be designed later.

10. Programmer Priorities

Programming should prove game logic before presentation polish. Placeholder UI and numbers are acceptable until the core loop is functional.

Priority	System	Minimum Proof
1	Branching Dialogue	Choices branch correctly and can evaluate else-if conditions.
2	Game Memory / Flags	Past decisions, context discoveries, approval, and story states can be stored and checked later.
3	Attributes / Character Build	CHA/INT/DEX can be allocated, modified by Class, stored, displayed, and referenced by checks.
4	D20 Resolution	D20 + derived modifier versus preset DC; four outcome states branch correctly.
5	Action Slots	Activities consume available slots; slots can refresh by day/period; unavailable actions are blocked.
6	Money	RM can be gained/spent/checked; daily RM5 allowance can be granted; purchases can fail if unaffordable.
7	Best Friend System	Track 3 companions, approval, active companion, passive buff, and contextual dialogue conditions.
8	Subject/Minigame Hooks	A generic way for minigames to return success/score/result and apply buffs/effects.
9	3D Environment / Device Motion	Research and prototype gyro camera vs motion/gyro parallax; determine simplest stable implementation.
10	Tap/Click Investigation	Interactive background zones can trigger flags, items, dialogue, or scripted events.
Low	Walkable Hub	Do not prioritize unless all core systems are proven and schedule allows.

Required Condition Checks

The dialogue/event system should eventually be able to query combinations of:

- Previous decisions and Boolean flags
- Attribute values and Class
- D20 outcome
- Money amount
- Action Slots remaining
- Best Friend recruited/active state
- Best Friend approval
- Items/context discoveries
- Current day / time window
- Minigame result or temporary food buff

11. Prototype / Vertical Slice Scope

11.1 Vertical Slice Definition

[LOCKED] The current prototype is a complete 30-day school/exam slice rather than a single isolated scene.

The vertical slice should demonstrate the full rhythm of the intended game at reduced content scale: character creation, classes, attributes, reactive school incidents, subject minigames, recess economy, Action Slots, weekend work/study/story decisions, Best Friend approval and buffs, D20 checks, occasional Investigation Scenes, and the final exam.

11.2 Minimum Feature Checklist

- ☐ Character creation: name, gender, appearance preset, Class, 12-point attribute allocation
- ☐ 3 Classes and Class modifiers
- ☐ CHA / INT / DEX with derived D20 modifiers
- ☐ Branching dialogue + game-state memory
- ☐ Four-state D20 check resolution
- ☐ Math, English, Science minigame hooks (final mechanics TBD)
- ☐ Recess shop + RM5 daily allowance + food buffs
- ☐ Action Slot scheduling
- ☐ Weekend part-time work and study/story choices
- ☐ 3 Best Friends, 1 active, approval + passive/contextual benefits
- ☐ Random/reactive school incident system
- ☐ Scheduled story beat support
- ☐ Ambient gyro/parallax prototype
- ☐ At least one functional Investigation Scene
- ☐ Day progression through Day 30
- ☐ Final exam/outcome state
- ☐ Save/load suitable for intentional reloading

12. Failure and Consequence Philosophy

[LOCKED] Backbenchers uses fail-forward design. Failed checks should usually create content rather than stop content.

Possible failure consequences include detention, embarrassment, reduced approval, lost RM, lost item/reward, a harder later situation, a missed opportunity, an alternate scene, or a weaker preparation state. The final exam can also produce a failure outcome without requiring a traditional Game Over.

Critical Success should be rewarding but not mandatory to understand the story. Critical Failure should be memorable without routinely making a run unwinnable.

13. Scope, Risks and Production Controls

Risk	Why It Matters	Control
Branch Explosion	30 days × classes × attributes × companions can multiply writing quickly.	Use local consequences for small incidents; reserve persistent branching for major beats.
3D/Gyro Scope	Interactive environments can become expensive to author.	Ambient movement by default; Investigation Scenes only when gameplay needs them.
Minigame Overproduction	Three subjects plus events can create too many bespoke mechanics.	Build reusable minigame frameworks and vary parameters/content.
Economy Imbalance	Food buffs/part-time work could trivialize school challenges.	Keep RM values small; tune strong buffs against time cost and exam pressure.
Best Friend Content Cost	Three companions can triple dialogue permutations.	Only write unique companion responses when context meaningfully matches.
Random Incident Fatigue	Too many forced rolls can feel repetitive.	Cap frequency and mix pure decisions, rolls, jokes, items, and no-check scenes.
Vertical PC UX	Vertical layout can feel narrow on desktop.	Design intentional framing/background treatment rather than stretching the mobile canvas.

Production Rule

When schedule pressure occurs, preserve dialogue branching, state memory, attributes, D20 checks, school-day structure, Action Slots, Money, and Best Friend logic first. Reduce hub scope, gyro complexity, investigation frequency, bespoke minigames, and UI animation before cutting the core role-playing logic.

14. Open Decisions / Next Design Pass

The following areas are intentionally unresolved and should be tackled after this v0.1 GDD:

1. Main story premise beyond the transfer/exam setup and day-by-day story beats
2. Names, personalities, roles, approval arcs, and buffs for the 3 Best Friends
3. Exact Math, English, and Science minigames
4. Exam scoring/pass/fail system
5. Action Slot counts and weekday/weekend schedules
6. Food menu, prices, buffs, and item/storage rules
7. Part-time job options and payout values
8. Random school incident probability, cap, and event pool
9. Attribute cap at character creation and final D20 balance table/DC bands
10. Critical outcome confirmation (natural 1 / natural 20)
11. Exact gyro/motion-parallax implementation after technical research
12. UI/UX flow and screen specifications
13. School name, geographic/cultural specificity, uniforms, calendar, and environmental art bible
14. Prototype ending variations and what data is shown in the end-of-run summary